

ANALYSTLAB AFRICA — DATA SCIENCE INTERNSHIP

FinTrust Digital Bank Project | Batch E

Week 3 Submission - Model Development, Refinement & Validation

Data Science Track: FinTrust Risk Review Predictive Model - Model Development & Validation

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Data Science Track | AnalystLab Africa Experience Lab Internship Programme

Project: FinTrust Financial Intelligence & Digital Banking Support Solution

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1. Executive Summary

Week 3 continues the same FinTrust modelling project rather than starting a new analysis. The central question was whether the performance ceiling observed in Weeks 1 and 2 was primarily a limitation of the Week 2 model specification or a limitation of the information available in the supplied synthetic data. The work therefore combined feature refinement, additional nonlinear model families, hyperparameter search, threshold evaluation, error analysis and model interpretation while preserving the original split and success criteria.

The headline finding is that greater model complexity did not materially improve discrimination. Feature refinement reduced the model from 24 to 16 raw inputs while preserving cross-validated performance and correcting the Week 2 coefficient-sign problem. Gradient Boosting, HistGradientBoosting and a tuned Random Forest were then evaluated as challengers. None met the pre-registered replacement rule, so the refined logistic regression is retained as the candidate final model for Week 4.

Result	Value	Against project criterion
Refined logistic regression ROC-AUC (5-fold CV)	0.663 ± 0.021	Below the ≥ 0.75 target
Validation ROC-AUC	0.683	Highest Week 3 validation AUC
Recall at tuned threshold 0.4869	0.703	Meets the ≥ 0.70 binding constraint
Precision at that threshold	0.286	Explicit review-workload trade-off
Feature-set reduction	24 $\rightarrow$ 16 raw inputs	Interpretability improved with negligible CV change
Best new-model validation AUC	0.672 — Gradient Boosting	Does not replace incumbent
Candidate for Week 4	Refined Logistic Regression	Frozen before final test scoring

Five findings shape the Week 4 handoff:

- The Week 2 performance ceiling persists. The strongest nonlinear challenger, Gradient Boosting, reached validation ROC-AUC 0.672, below the refined logistic regression at 0.683.
- Feature refinement solved a genuine interpretability defect without buying performance through leakage or post-hoc target chasing. CV ROC-AUC moved only from 0.6650 ± 0.0174 to 0.6630 ± 0.0208 , while Amount_Band_Q5 changed to the directionally coherent +0.407 coefficient.
- The binding recall constraint remains achievable. At threshold 0.4869, validation recall is 0.703 and precision is 0.286.
- The error pattern remains uneven by transaction type. Transfer recall is 0.950 while Bill Payment recall is 0.269, so aggregate recall cannot be treated as uniform performance.
- The untouched test set remains reserved for the single Week 4 score. Week 3 therefore ends with a frozen candidate rather than another round of validation-driven model changes.

Responsible-use notice: Risk_Review_Flag is synthetic and is not a real fraud label. All results are educational evidence for review prioritisation only.

2. Applying the Week 2 Plan

This section mirrors the Week 2 report's "Applying the previous week's plan" structure. Each Week 2 action is traced to what was implemented, what changed and why.

2.1 Planned versus implemented

Week 2 plan	Status	Week 3 implementation / evidence
Reduce redundant amount encodings	Complete	Kept Amount_Band; removed Amount_Log and Is_High_Value. CV changed by only -0.0020 and Q5 coefficient became +0.407.
Test whether Amount vs Cust Mean adds independent value	Complete	Removed after Week 2 permutation importance of only 0.003 and evidence of confounding with absolute amount.
Remove weak temporal/activity features	Complete	Transaction_DayOfWeek, Is_Weekend and Cust_Txn_Count removed.
Review weak fairness-sensitive candidates	Complete	Age and Monthly_Income_Band removed; Gender remains excluded.
Test stronger nonlinear algorithms	Complete	Gradient Boosting and HistGradientBoosting added; tuned Random Forest also evaluated.
Hyperparameter tuning	Complete	RandomizedSearchCV used on training folds; validation reserved for comparison/threshold selection.
Deepen error analysis	Complete	False positives/false negatives and transaction-type patterns quantified.
Prepare a Week 4 candidate	Complete	Refined Logistic Regression frozen; test set remains untouched.

2.2 What changed from the Week 2 approach, and why

Change	Week 2 position	Week 3 position	Reason
Amount representation	Amount_Log, Amount_Band and Is_High_Value all present	Amount_Band retained alone	Three overlapping encodings destabilised coefficients;

Customer-relative amount	Amount vs Cust_Mean retained	Removed	Amount_Band had the strongest Week 2 amount importance. Week 2 importance only 0.003; support for H1 did not translate into independent model value.
Model families	LR, Decision Tree, Random Forest	Adds tuned RF, Gradient Boosting and HistGradientBoosting	Tests whether the ceiling is algorithmic rather than data-driven.
XGBoost	Planned candidate	Not used	Unavailable in the execution environment; sklearn histogram boosting supplied a justified tree-boosting alternative.
Candidate interpretation	Permutation importance required because coefficients conflicted	Permutation importance retained; coefficients again usable as a secondary check	Feature reduction corrected the multicollinearity-driven sign problem.

No change was made to the target definition, the 70/15/15 stratified split, seed 42, primary metric, recall constraint, replacement rule or test-set policy. Keeping these commitments fixed is essential to making Week 3 a genuine validation exercise rather than a moving-target optimisation exercise.

3. Resources and Tools Used

3.1 Resources

Resource	How it was used in Week 3	What it provided	Limitations encountered
FinTrust_Transaction_Data.csv	Primary transaction-level source and synthetic target	Amount, time, transaction type, channel, location, international status and Risk_Review_Flag	Synthetic target definition remains undocumented; no velocity, counterparty, device-fingerprint or network signals.
FinTrust_Customer_Data.csv	Customer attributes joined by Customer_ID	Segment, account, tenure and engagement context	Customer-level attributes remain weak predictors.
Week 2 notebook and modelling outputs	Exact methodological continuation	Split logic, baseline specification, feature evidence, permutation importance and risk register	Week 2 feature redundancy had to be corrected before coefficients could be interpreted safely.
Week 2 prepared modelling dataset	Reproducibility/reference check	Confirmed previous engineered-feature definitions and dimensions	Not used as a shortcut around training-only fitting; Week 3 reconstruction preserves the original split logic.
FinTrust project brief / roadmap	Business framing and scope control	Review-prioritisation framing and four-week sequence	No quantified business cost ratio for false positives versus false negatives.

3.2 Tools

Tool	What it was used for	Why appropriate
Python 3 / Jupyter / Colab	End-to-end reproducible Week 3 analysis	Keeps code, outputs and reasoning together; notebook verified to run end-to-end.
pandas / NumPy	Data preparation, feature transformations, result tables	Consistent with Week 2 and sufficient for tabular synthetic data.
scikit-learn Pipeline / ColumnTransformer	Preprocessing, fitting and fold-safe transformation	Refits preprocessing inside each CV fold, enforcing leakage control mechanically.
RandomizedSearchCV	Bounded tuning of tree/boosting candidates	Tests useful parameter combinations without an excessive exhaustive grid.
GradientBoostingClassifier	First additional model family	Tests sequential nonlinear boosting using standard sklearn.
HistGradientBoostingClassifier	Second additional model family	Efficient histogram-based nonlinear boosting; justified substitute when XGBoost was unavailable.
Permutation importance	Validation-set model interpretation	Model-agnostic and directly comparable with Week 2 interpretation.
Git / GitHub	Version control and portfolio submission	Preserves the Week 1 → Week 2 → Week 3 progression in one repository.

4. Part A — Baseline Model Review

Week 3 begins from the selected Week 2 logistic-regression baseline, not from a new specification. The review covers performance, features, preprocessing, metrics, error behaviour and limitations, as required by Part A.

4.1 Performance carried forward from Week 2

Measure	Week 2 evidence	Week 3 implication
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5-fold CV ROC-AUC	0.665 ± 0.017	Below the ≥0.75 target; meaningful challenger must clear the replacement rule.
Validation ROC-AUC	0.674	Modest discrimination; room for improvement.
Recall at default 0.5	0.671	Below binding ≥0.70 constraint.
Recall at tuned threshold	0.703	Constraint achievable through threshold selection.
Precision at tuned threshold	0.283	High review workload must be reported explicitly.
Top-decile lift	1.96×	Some prioritisation value despite modest AUC.

4.2 Feature-set review

Week 2 interpretation identified a clear hierarchy: Transaction_Type and Is_Overnight dominated; amount and international status added secondary signal; customer-level attributes contributed almost nothing. The major defect was not leakage but redundancy: Amount_Log, Amount_Band and Is_High_Value encoded the same underlying quantity, while Amount_vs_Cust_Mean added little once absolute amount was present.

4.3 Preprocessing review

The preprocessing design itself was retained. Numeric scaling and categorical one-hot encoding remain inside a ColumnTransformer embedded in each scikit-learn Pipeline. This is important because cross-validation refits preprocessing inside every training fold. The Week 3 improvement therefore targets feature content and model family, not the leakage-safe architecture.

4.4 Areas requiring improvement

- Remove redundant and near-zero-value features so the candidate can be interpreted directly.
- Test at least two additional nonlinear classifiers beyond the Week 2 LR/DT/RF set.
- Tune tree-based candidates without using the test set.
- Re-run threshold selection against the binding recall constraint for every candidate.
- Go beyond aggregate recall and quantify false-negative/false-positive patterns by transaction type.
- Freeze the selected candidate after Week 3 to reduce validation overfitting before Week 4.

5. Part B — Additional Model Development

Three challengers were evaluated on the refined Week 3 feature set: a tuned Random Forest and two additional model families, Gradient Boosting and HistGradientBoosting. The two boosting models satisfy the assignment requirement for at least two classifiers beyond the three Week 2 models.

5.1 Search discipline

Hyperparameter search used training-fold ROC-AUC. Validation was not used to fit preprocessing or search parameters; it was used only after fitting for model comparison and threshold selection. The test split remained untouched.

Model	Best CV ROC-AUC	Validation ROC-AUC	Role / finding
Tuned Random Forest	0.662	0.668	Checks whether tuning rescues the Week 2 ensemble; it does not.
Gradient Boosting	0.664	0.672	Best new nonlinear challenger, but below the refined LR on validation.
HistGradientBoosting	0.658	0.667	Second required new family; no material gain.

5.2 Why XGBoost was not used

XGBoost was planned as a possible Week 3 candidate but was unavailable in the execution environment and could not be installed there. The substitution is documented rather than hidden: scikit-learn's GradientBoostingClassifier and HistGradientBoostingClassifier provide two legitimate gradient-boosting approaches without adding an unverified external dependency.

5.3 Result of the algorithmic-ceiling test

The best new challenger reached validation ROC-AUC 0.672. The refined logistic regression reached 0.683. More importantly, no challenger exceeded the incumbent by more than the incumbent's own cross-fold standard deviation. The pre-registered replacement rule therefore does not trigger. The result supports the Week 2 narrative that the main constraint is the feature information available, not simply the absence of a more sophisticated classifier.

6. Part C — Feature Refinement

Feature refinement was evidence-led. No feature was removed because it was inconvenient to the result; each decision traces to Week 2 importance, hypothesis evidence, fairness scrutiny or the multicollinearity defect.

Feature decision	Reason	Expected effect	Observed result
Keep Amount_Band; drop Amount_Log and Is_High_Value	Three encodings represented the same signal and destabilised coefficients	Reduce multicollinearity without material information loss	Q5 coefficient becomes +0.407; CV change negligible.
Drop Amount_vs_Cust_Mean	Week 2 permutation importance only 0.003; confounded with absolute amount	Simpler amount interpretation	No material performance loss.
Drop Transaction_DayOfWeek and Is_Weekend	No useful Week 2 signal	Reduce noise/dimensionality	Refined model remains within baseline CV noise.
Drop Cust_Txn_Count	Near-zero importance and weak behavioural meaning	Remove weak feature	No material loss.
Drop Age and Monthly_Income_Band	Negligible signal under fairness scrutiny	Avoid weak demographic/proxy candidates	Candidate uses fewer fairness-sensitive inputs.
Continue excluding Gender	Protected attribute; prior fairness commitment	Prevent direct protected-attribute use	Exclusion maintained.

6.1 Re-test after refinement

Specification	Raw inputs	5-fold CV ROC-AUC	Interpretability
Week 2 logistic regression	24	0.6650 ± 0.0174	Amount-band coefficient signs unstable
Week 3 refined logistic regression	16	0.6630 ± 0.0208	Amount_Band_Q5 +0.407; direction coherent

The -0.0020 change in mean CV ROC-AUC is negligible relative to cross-fold variability. The refinement therefore succeeds on its intended objective: remove redundant/weak inputs and restore direct interpretability without sacrificing meaningful predictive information.

7. Part D — Model Comparison

Every model is compared at a validation threshold selected to satisfy recall ≥ 0.70 . Accuracy is shown because the assignment asks for it, but it remains excluded from selection for the same reason established in Week 1: with a 19.6% positive class, accuracy can reward models that simply avoid predicting the minority class.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Threshold
Logistic Regression	0.599	0.286	0.703	0.407	0.683	0.4869
Decision Tree	0.551	0.262	0.708	0.382	0.633	0.5063
Random Forest	0.569	0.270	0.703	0.390	0.658	0.3689
Tuned Random Forest	0.574	0.273	0.703	0.393	0.668	0.4692
Gradient Boosting	0.565	0.269	0.708	0.389	0.672	0.1913
HistGradientBoosting	0.582	0.277	0.703	0.397	0.667	0.2072

7.1 Trade-off discussion

- Logistic regression has the highest validation ROC-AUC (0.683) and the highest F1 (0.407) among the compared recall-constrained operating points.
- Decision Tree and Gradient Boosting obtain slightly higher recall (0.708), but that small gain is accompanied by lower precision and substantially lower ROC-AUC.
- HistGradientBoosting produces the highest accuracy among the challengers (0.582) and precision 0.277, but neither changes the candidate decision because accuracy is not a selection metric and its ROC-AUC is only 0.667.
- The tuned Random Forest improves on the untuned forest's validation ROC-AUC from 0.658 to 0.668, but the improvement is insufficient to replace logistic regression.
- No model reaches the project ROC-AUC target of 0.75. That shortfall remains a reported result, not a criterion to tune away.

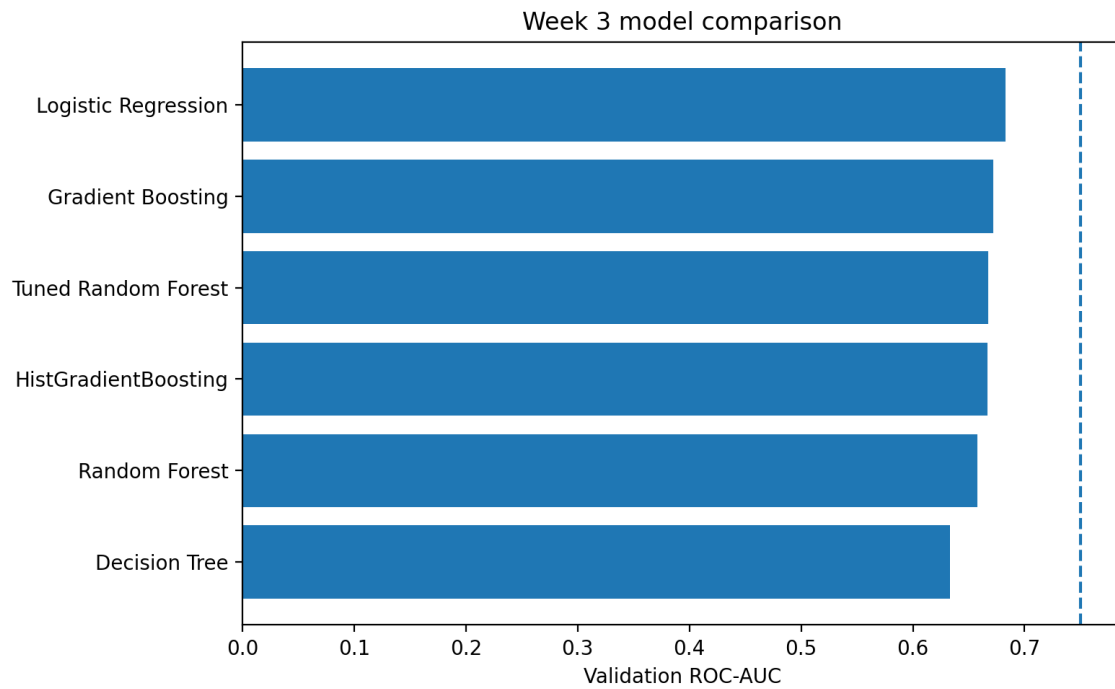


Figure 15 — Week 3 validation ROC-AUC comparison.

8. Part E — Error Analysis

At the candidate threshold of 0.4869, the refined logistic regression produces $TN=830$, $FP=618$, $FN=105$ and $TP=248$ on validation. The 105 false negatives are synthetic review cases that would not be prioritised; the 618 false positives represent additional analyst workload. The model therefore meets recall by accepting low precision rather than by creating an impression of uniformly strong classification.

Outcome	Count	Operational reading
True negatives	830	Transactions correctly left outside the prioritised review queue.
False positives	618	Unflagged synthetic cases sent to review; main workload cost.
False negatives	105	Flagged synthetic cases missed by the queue; binding risk controlled through recall threshold.
True positives	248	Flagged synthetic cases correctly prioritised.

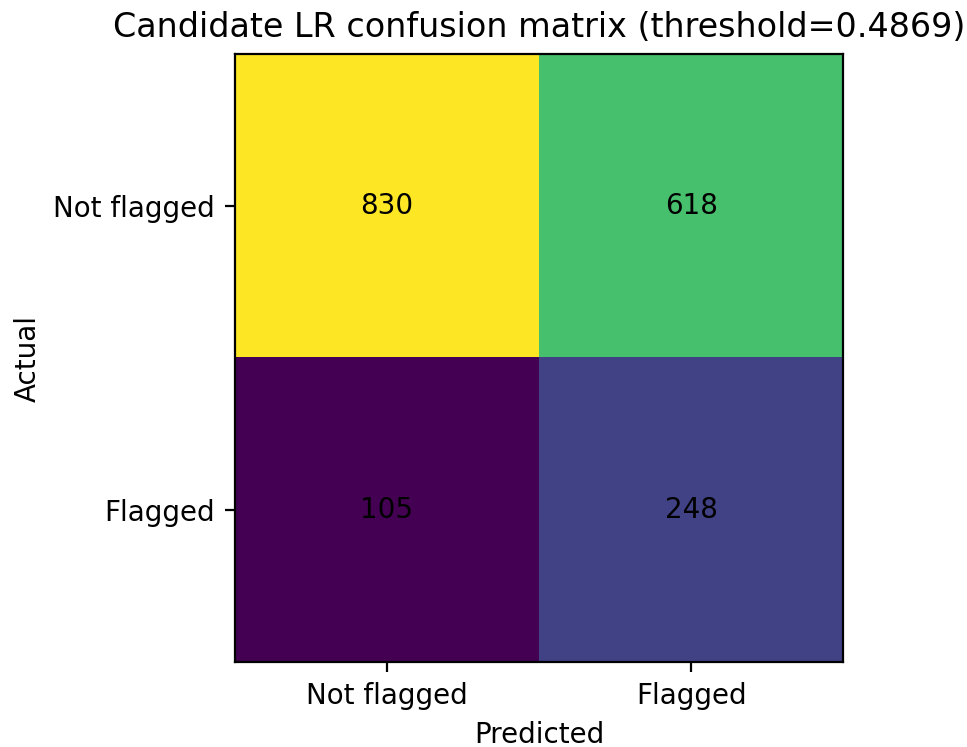


Figure 16 — Candidate logistic-regression confusion matrix at the recall-constrained validation threshold.

8.1 Error patterns by transaction type

Transaction type	n	Recall	Precision	FP	FN
Airtime/Data	166	0.476	0.200	40	11
Bill Payment	209	0.269	0.259	20	19
Card Purchase	470	0.311	0.271	51	42
Cash Withdrawal	193	0.885	0.289	113	6
Deposit	202	0.424	0.250	42	19
Transfer	561	0.950	0.302	352	8

8.2 What the errors mean

The failure mode is systematic. Transfer recall is 0.950 and Cash Withdrawal recall is 0.885, while Bill Payment is only 0.269 and Card Purchase 0.311. This means the model is effective where the supplied features strongly identify high-base-rate transaction categories, but it has little information for distinguishing flagged from unflagged cases inside low-signal categories. The error analysis therefore reinforces, rather than contradicts, the performance-ceiling conclusion.

The model should not be described as uniformly detecting risky transactions. A more accurate statement is that it prioritises the synthetic review label unevenly, with strong coverage in transfers/cash withdrawals and weak coverage in bill payments/card purchases.

9. Part F — Model Interpretation

Permutation importance remains the primary interpretation method so that Week 3 is directly comparable with Week 2. Logistic-regression coefficients are now added as a secondary directional check because feature refinement removed the amount multicollinearity defect.

Feature	Permutation AUC drop	Interpretation
Transaction_Type	0.084	Dominant signal; transaction category drives much of the ranking.
Is_Overnight	0.052	Strong time-of-day signal consistent with Week 2.
Amount_Band	0.018	Upper-tail amount effect remains useful after redundant encodings are removed.
Is_International	0.011	Strong per-case effect but limited by low prevalence.
Channel	0.004	Weak incremental contribution.
Account_Type	0.004	Weak customer-level contribution.
Tenure_Band	0.004	Weak; consistent with rejected tenure hypothesis.
Engagement_Band	0.003	Weak customer-level contribution.

Location	0.002	Near-noise contribution.
Tenure_Months	0.001	Near-noise contribution.

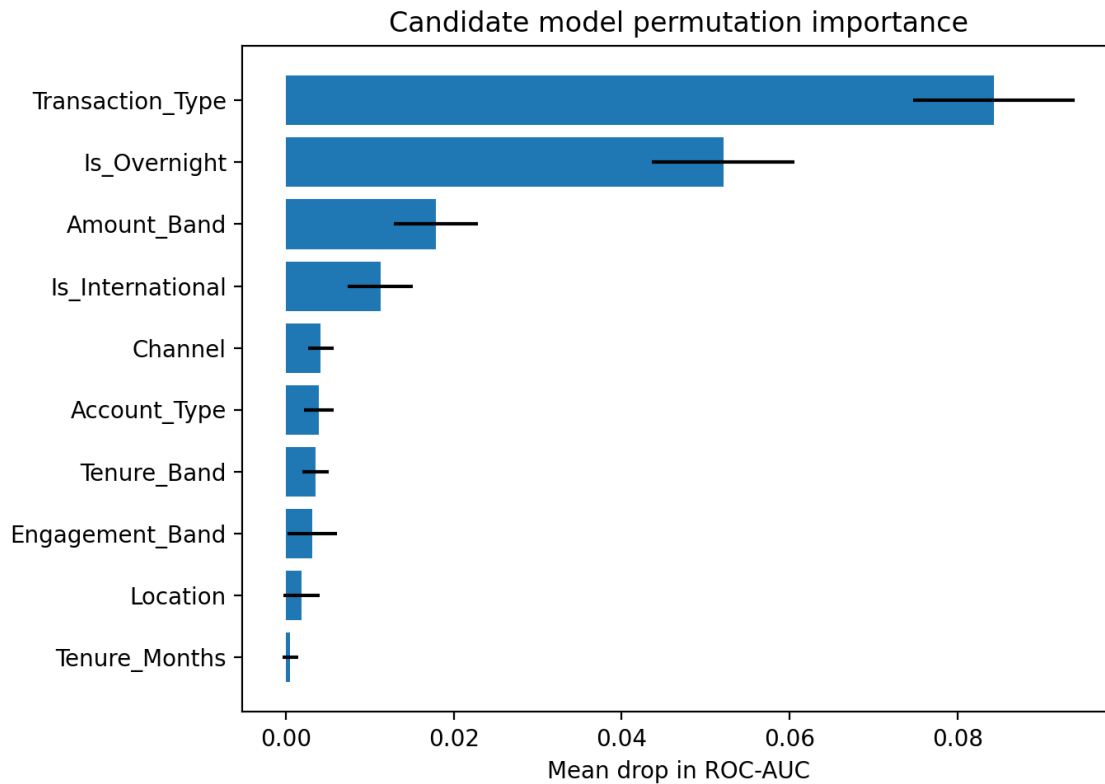


Figure 17 — Candidate model permutation importance on validation data.

9.1 Coefficient sanity check

Term	Coefficient	Reading
Is_International	+0.942	International transactions receive higher model scores, consistent with training evidence.
Is_Overnight	+0.830	Overnight activity increases the model score.
Transaction_Type_Transfer	+0.672	Transfers are associated with higher review priority.
Transaction_Type_Cash Withdrawal	+0.596	Cash withdrawals are associated with higher review priority.
Amount_Band_Q5	+0.407	Top amount band now has the expected positive direction.

The key validation result is Amount_Band_Q5. In Week 2 its sign contradicted the descriptive evidence because three amount encodings competed inside the same linear model. After refinement it is +0.407, matching the observed high-value pattern. This closes the Week 2 interpretation defect rather than merely working around it.

10. Part G — Candidate Final Model

The candidate final model is frozen at the end of Week 3 so that Week 4 can perform one final untouched test evaluation rather than continuing to optimise against validation.

Item	Decision / evidence
Model	Refined Logistic Regression
Features used	16 raw inputs after Week 3 refinement
Preprocessing	StandardScaler for numeric inputs and OneHotEncoder for categoricals inside ColumnTransformer/Pipeline
Training/validation design	Original seed-42 stratified 70/15/15 split; training-only fitting; validation for Week 3 comparison
5-fold CV ROC-AUC	0.663 ± 0.021
Validation ROC-AUC	0.683
Operating threshold	0.4869
Recall / precision	0.703 / 0.286
Why selected	Highest validation AUC, interpretable, meets recall constraint, and no challenger cleared replacement rule
Known limitation	ROC-AUC target 0.75 remains unmet; low precision and missing behavioural feature categories

Week 4 status

Candidate frozen; test set not yet scored

10.1 Why the simpler model wins

The selection is not a preference for simplicity in the abstract. It follows the pre-registered evidence rule. A more complex model would replace the incumbent only if its improvement exceeded the incumbent's own cross-fold standard deviation. None did. Retaining logistic regression therefore applies the same rule that prevented the Week 2 Random Forest from replacing it on a marginal raw AUC difference.

10.2 What would be needed to move beyond the ceiling

The evidence increasingly points toward missing feature categories rather than missing algorithm complexity. Useful future categories would include transaction velocity, device fingerprint consistency, counterparty/network relationships and richer geolocation behaviour. Those variables are not present in the supplied synthetic dataset, so Week 3 does not invent proxies for them or claim they were tested.

11. Mandatory Validation Chain

The assignment requires the explicit chain Initial Output → Test → Finding → Refinement → Re-test → Result. Week 3 contains two linked chains: one addressing interpretability and one addressing the performance ceiling.

Initial Output	Test	Finding	Refinement	Re-test	Result
Week 2 LR: CV AUC 0.665; contradictory amount coefficient	Feature redundancy review	Three overlapping amount encodings caused unstable interpretation	Keep Amount_Band; remove redundant/weak inputs	5-fold CV + coefficient inspection	CV AUC 0.663 ± 0.021; Q5 coefficient +0.407
Week 2 performance ceiling	Tuned RF + GB + HGB	No challenger produced material CV gain	Retain simpler candidate	Validation AUC + replacement rule	LR validation AUC 0.683; ceiling persists

11.1 Improvement demonstrated

The meaningful improvement is not an inflated AUC claim. It is a model-design improvement: the refined specification removes eight weak/redundant raw inputs, restores directionally coherent coefficients and preserves performance within cross-fold noise. The re-test demonstrates that the Week 2 interpretability problem was caused by feature redundancy rather than by information that had to be retained for discrimination.

12. Testing and Quality Assurance

This section follows the same prescribed Week 2 structure: Work Completed → Test/Evaluation → Finding → Action/Improvement → Result.

#	Work Completed	Test / Evaluation	Finding	Action / Improvement	Result
T1	Exact split reconstruction	Checked row counts and class rates	8,399 / 1,801 / 1,800 structure preserved	Reuse seed-42 split logic	Week 2–3 evidence remains comparable
T2	Feature refinement	5-fold stratified CV	Mean AUC changed by only -0.0020	Retain reduced set	Interpretability improved without material loss
T3	Coefficient interpretation	Compared refined signs with descriptive evidence	Amount_Band_Q5 now +0.407	Use coefficients as secondary check	Week 2 sign bug resolved
T4	Additional model development	RandomizedSearchCV + validation comparison	No challenger materially replaces LR	Keep incumbent	Replacement rule applied literally
T5	Recall constraint	Validation threshold search	Default operating point insufficient for binding rule	Tune threshold to 0.4869	Recall 0.703
T6	Error analysis	Confusion matrix + transaction-type breakdown	Recall highly uneven by type	Carry limitation into candidate decision	Aggregate recall not overstated
T7	Interpretation	Permutation importance + coefficients	Transaction-level features dominate	Report both model-agnostic and directional evidence	Week 1–2 signal hierarchy confirmed
T8	Test-set protection	Checked Week 3 workflow for test scoring	No Week 3 test evaluation performed	Freeze candidate	Single Week 4 scoring remains valid

13. Decision Log

Week 3 continues the Week 2 decision numbering rather than restarting it.

Decision	Reason	Evidence	Impact
D9 — Reduce feature set	Remove redundancy and weak inputs	CV -0.0020 only; Q5 sign fixed	Simpler, more interpretable candidate
D10 — Use sklearn GB/HGB	XGBoost unavailable in execution environment	No install/network access; sklearn alternatives available	Two legitimate additional model families delivered
D11 — Retain Logistic Regression	Replacement rule not met	Best new validation AUC 0.672 vs LR 0.683; no material CV gain	Candidate frozen for Week 4
D12 — Tune operating threshold	Recall ≥ 0.70 is binding	Threshold 0.4869 gives recall 0.703	Accept precision 0.286 as explicit workload trade-off
D13 — Freeze after Week 3	Repeated validation comparison creates overfitting risk	Multiple models already compared on validation	No further candidate changes before test scoring
D14 — Keep permutation importance as primary interpretation	Maintains continuity and is model-agnostic	Week 2 and Week 3 importance rankings directly comparable	Coefficients used only as secondary directional check

14. Assumptions, Limitations and Risks — Week 2 Review

The Week 2 risk register is carried forward rather than rewritten. Status changes are evidence-based and new Week 3 risks are added separately.

14.1 Prior risk register review

Risk	Week 2 status	Week 3 status	Evidence / action
R1 — Undocumented synthetic target	Materialised	Active	Target provenance still unavailable; non-fraud disclaimer remains mandatory.
R2 — Transaction Status leakage	Resolved	Resolved	Field remains excluded.
R3 — Class imbalance	Controlled	Controlled	Threshold tuning continues to enforce recall constraint; accuracy excluded from selection.
R4 — Repeated-customer dependence	Resolved	Resolved	Week 2 grouped sensitivity was negligible; no contrary Week 3 evidence.
R5 — Performance ceiling	Materialised / managed	Materialised	Boosting did not materially improve ROC-AUC.
R6 — Final-test overfitting	Not yet active	Controlled / not yet active	Test set remains untouched; candidate frozen.
R7 — Synthetic-data generalisability	Active	Active	No real-world validation; all claims remain educational.
R8 — Review-cost assumptions	Active	Active	No supplied FP/FN cost ratio; recall-weighted threshold remains an assumption.
R10 — Temporal realism	Active	Active	Synthetic temporal generation limits behavioural/seasonality inference.
R11 — Governance / target provenance	Active	Active	Human oversight and documented provenance would be required operationally.
R12 — Feature redundancy	Active	Resolved	Redundant amount encodings removed; coefficient direction corrected.
R13 — Uneven segment recall	Active	Active	Week 3 error analysis confirms recall from 0.269 to 0.950 across types.
R14 — International coefficient stability	Active	Active	Important signal but limited prevalence; no new data added.

14.2 New Week 3 risks

New risk	Status	Why it matters	Mitigation
R15 — Validation overfitting from repeated model comparison	Active	Week 3 compares several candidates on the same validation split	Freeze candidate now; score test once in Week 4.
R16 — Low precision at recall-constrained threshold	Active	Precision 0.286 implies substantial review workload	Report workload explicitly; do not claim automated decisioning.
R17 — Missing behavioural feature categories	Materialised limitation	Model complexity did not close the AUC gap	Document absent velocity/device/network/geolocation signals; do not fabricate them.

14.3 Model and data limitations

- Validation ROC-AUC 0.683 remains below the pre-set 0.75 target.
- Precision 0.286 at the recall-constrained threshold means many reviewed cases would not carry the synthetic flag.
- Recall varies materially by transaction type; the model is not uniformly effective across products.
- Risk_Review_Flag is synthetic and undocumented; it cannot be interpreted as fraud truth.
- The available data omit behavioural categories that could plausibly improve within-category discrimination.

- No Week 3 result establishes real-world generalisability, fairness or operational suitability.

15. Week 3 Project Summary

The table below answers the Week 3 assignment points directly and leaves a clear Week 4 handoff.

Assignment point	Week 3 answer	Evidence location
Review baseline model	Week 2 LR reviewed for performance, preprocessing, feature weaknesses, errors and limitations	§4
Train ≥ 2 additional classifiers	Gradient Boosting and HistGradientBoosting added; tuned RF also evaluated	§5
Feature refinement	24 raw inputs reduced to 16; redundant/weak/fairness-sensitive candidates removed	§6
Model comparison	All six models compared on accuracy, precision, recall, F1 and ROC-AUC	§7
Error analysis	618 FP, 105 FN, 248 TP at candidate threshold; type-level patterns quantified	§8
Model interpretation	Permutation importance plus corrected LR coefficients	§9
Candidate final model	Refined Logistic Regression, threshold 0.4869, validation AUC 0.683	§10
Validation chain	Feature-redundancy fix and algorithmic-ceiling test both re-tested	§11
Testing / QA	Eight explicit tests documented in required structure	§12
Limitations / risks	Prior register updated; R15–R17 added	§14
Week 4 readiness	Candidate frozen; test set untouched for one final score	§10, §14

15.1 Week 4 entry point

- Restart and run the final notebook in a clean environment to confirm reproducibility.
- Load the frozen refined logistic-regression specification without further validation-driven tuning.
- Score the untouched test split once and report ROC-AUC plus the pre-defined supporting metrics.
- Perform final fairness/segment review and document any gap between validation and test performance.
- Prepare final documentation and presentation without reframing Risk_Review_Flag as real fraud detection.

16. Appendices

Appendix A — File manifest

Artifact	Purpose
notebooks/03_week3_model_development_and_validation.ipynb	Self-contained Week 3 analysis; Google Drive data loading; model development, validation and interpretation.
reports/week3/FinTrust_Week3_DataScience_Submission_TJGumbo.docx	Editable Week 3 report.
reports/week3/FinTrust_Week3_DataScience_Submission_TJGumbo.pdf	Submission-ready Week 3 report.
reports/figures/fig15–fig17	Week 3 model comparison, confusion matrix and permutation-importance figures.
README.md	Current project results, weekly progress and limitations.
WEEK3_SUBMISSION_CHECKLIST.md	Assignment-to-evidence mapping and submission checks.
social_posts_week3.md	LinkedIn and optional X copy for Week 3.

Appendix B — Reproducibility statement

The analysis uses seed 42 and the original stratified 70/15/15 split created before exploratory analysis or feature fitting. All fitted statistics are learned from training data only. Preprocessing is embedded inside scikit-learn Pipeline/ColumnTransformer objects so cross-validation refits transformations per fold. Validation is used for Week 3 model comparison, threshold selection and interpretation. The test set is not scored in Week 3 and is reserved for one final Week 4 evaluation.

Appendix C — Methodological commitments retained

- Train/validation/test split fixed before EDA or feature engineering: 70/15/15, stratified, seed 42.
- All EDA and fitted statistics use the training split only.
- Test set remains untouched until Week 4.
- Preprocessing remains inside Pipeline/ColumnTransformer objects.
- Primary metric remains ROC-AUC with recall ≥ 0.70 as the binding constraint.
- Accuracy is reported but excluded from model selection.
- Replacement rule remains literal: a challenger must beat the incumbent by more than the incumbent cross-fold standard deviation.
- Rejected hypotheses remain reported as results.

- Gender remains excluded; weak demographic candidates were removed under fairness scrutiny.
- No result is claimed unless produced by executed code.

Appendix D — Responsible-use statement

This project uses entirely synthetic educational data. Risk_Review_Flag is a synthetic review-prioritisation label and must not be interpreted as real fraud detection, a fraud probability, or an automated basis for adverse customer action. Model outputs are suitable only for educational demonstration of a review-prioritisation workflow. Human oversight, documented target provenance, representative real-world validation, fairness testing and governance would be required before any operational use.